SYNTHETIC DECEPTION AT SCALE

Deepfake Audio Detection

Technical Research Report

Evaluating model accuracy, cross-dataset generalization, feature representation, and deployment-domain failure in AI-generated speech detection

Ryan Cherukuri • Pedro Magalhaes Brito Scardazzi • Isaac Euceda-Colindres

Senior Capstone Project • Fall 2026

RESEARCH FOCUS

How well do deepfake-audio detectors generalize beyond the datasets on which they were trained?

Abstract

This project investigated the reliability of machine-learning approaches for detecting AI-generated or manipulated speech. The initial system used 20 averaged Mel-Frequency Cepstral Coefficients (MFCCs) and achieved strong held-out performance on DFADD, but accuracy collapsed when evaluated on independent datasets. We therefore compared multiple classifiers, balanced training across DFADD and ASVspoof, expanded the acoustic representation from 20 to 130 features, tested augmentation, evaluated a pretrained WavLM representation, and performed local microphone deployment tests. The final richer-feature multilayer perceptron (MLP) achieved 99.75% held-out accuracy, 69.5% on In-the-Wild, and 81.0% on an ASVspoof evaluation set. Despite these gains, fresh genuine microphone recordings were still misclassified as synthetic with very high confidence. The central finding is that benchmark accuracy alone is not sufficient evidence of deployment reliability: feature representation and domain diversity improved generalization, but domain shift remained the dominant challenge.

Headline results

99.75%	69.5%	81.0%	99%+
Held-out richer MLP	In-the-Wild	ASVspoof evaluation	Wrong confidence on real mic

Primary research finding
The project evolved from “Can we obtain high classification accuracy?” to “Does that accuracy survive a change in dataset, microphone, codec, or recording environment?” Cross-dataset testing exposed weaknesses that ordinary train/test splits did not.

1 Introduction

Synthetic speech can now imitate natural human voices closely enough that simple listening is often insufficient for reliable identification. This creates technical and social risks because fabricated audio can be used to impersonate individuals, manufacture evidence, or amplify misinformation. A detector therefore needs more than high benchmark accuracy; it must also perform consistently when audio comes from new speakers, devices, codecs, and synthesis systems.

Our software objective was to build an audio-classification pipeline that labels a recording as genuine or synthetic and provides a model confidence score. The research objective became more demanding: measure how model performance changes when the evaluation domain differs from the training domain, then identify which interventions actually improve generalization.

Research questions

- How well does a detector trained on DFADD generalize to independent datasets?
- Does changing the classifier solve cross-dataset failure?
- Does balancing training across multiple datasets improve robustness?
- Does a richer acoustic representation improve cross-domain performance?
- Can simple augmentation or pretrained WavLM embeddings close the remaining deployment gap?
- How trustworthy are model confidence scores on unfamiliar real-world recordings?



Figure 1. Final detection pipeline used for the deployed research prototype.

2 Data and Preprocessing

Datasets used

Dataset / source	Role in project	Class structure	Notes
DFADD (isjwdu/DFADD)	Primary training source	real / spoofed	~207,955 rows; ~42.7 GB; streamed rather than fully downloaded
ASVspoof 2019 LA	Secondary training + evaluation family	bonafide / spoof	Used to increase source diversity; evaluation split kept separate
In-the-Wild	Independent external evaluation	bonafide / spoof	Primary cross-dataset generalization test
WaveFake	Investigated only	primarily synthetic metadata in accessed copy	Not used in final model because of storage / balance constraints
Local QuickTime / macOS speech	Deployment stress tests	genuine / synthetic	Used qualitatively, not as a statistically powered test set

Streaming and class balance

DFADD was too large to download conveniently in full, so the project used Hugging Face streaming. We discovered that taking samples directly from the raw stream could preserve ordering and distort the class distribution. To reduce this risk, the stream was shuffled with a 10,000-example buffer before sampling. Later experiments explicitly enforced equal class counts and, where relevant, equal dataset-source counts.

Standard audio preprocessing

- Audio loaded as mono speech.
- Sample rate standardized to 16,000 Hz.
- Very short clips were rejected before feature extraction.
- For streamed datasets, decoding was performed manually when necessary to avoid codec/toolchain problems.
- Training and evaluation splits were kept separate; In-the-Wild was not used as training data for the primary generalization experiments.

3 Methodology

Baseline representation: 20 averaged MFCCs

The first controlled models represented each clip with 20 MFCC coefficients averaged over time. MFCCs summarize the spectral envelope of speech and are widely used in speech processing, but temporal averaging compresses each recording into a very small vector. The representation therefore discards much of the variation occurring throughout the clip.

Classifier experiments

To determine whether the original MLP architecture was responsible for poor external performance, we trained additional classifiers on the same basic feature space. These included Random Forest, Support Vector Machine, Gradient Boosting, Logistic Regression, Extra Trees, and K-Nearest Neighbors. This allowed us to separate “classifier choice” from “representation and data-domain” effects.

Multi-dataset training

We next combined DFADD with ASVspoof. The first combined model was class-balanced but still heavily dominated by DFADD. We then built an equal-source dataset containing 8,000 DFADD and 8,000 ASVspoof examples, with each source itself balanced between real and spoofed audio.

Richer 130-feature representation

The strongest feature-engineering intervention expanded each clip from 20 values to 130 acoustic statistics. The representation retained both central tendency and variation, and it incorporated short-term dynamics through delta features.

Feature group	Count	Purpose
MFCC means	20	Average spectral-envelope information
MFCC standard deviations	20	Within-clip spectral variability
Delta-MFCC means / std	40	First-order temporal change
Delta-delta MFCC means / std	40	Second-order temporal change
Spectral centroid, bandwidth, rolloff, zero-crossing, RMS (mean + std)	10	Additional frequency and energy statistics
Total	130	Final feature vector per recording

Augmentation and pretrained representation experiments

A separate robustness experiment applied gain changes, Gaussian noise, and resampling perturbations to generate augmented copies while using group-aware train/test splitting to prevent an original clip and its augmented copy from leaking across the split. Finally, a WavLM Base Plus pilot extracted 768-dimensional pretrained speech embeddings from 1,000 balanced DFADD/ASVspoof clips; those embeddings were classified with logistic regression.

4 Results

Balanced DFADD baseline

The first strong controlled baseline used 10,000 real and 10,000 spoofed DFADD recordings. A StandardScaler + MLP model achieved 95.65% accuracy on 4,000 held-out samples.

Actual class	Predicted real	Predicted spoof
Real	1904	96
Spoof	78	1922

Interpretation
The baseline appeared reliable inside DFADD, but this result measured performance within a familiar data distribution. It did not yet establish real-world generalization.

Cross-dataset collapse

The same baseline was then evaluated on 200 balanced In-the-Wild samples. Accuracy fell to 53.0%, and 194 of the 200 clips were predicted as spoofed. On 200 ASVspoof evaluation samples, the baseline achieved 50.0% and classified every sample as spoofed. These tests revealed a severe domain-shift problem.

Evaluation set	Accuracy	Confusion matrix summary
DFADD held-out	95.65%	[[1904, 96], [78, 1922]]
In-the-Wild	53.00%	[[6, 94], [0, 100]]
ASVspoof	50.00%	[[0, 100], [0, 100]]

Changing the classifier did not solve generalization

Model	Internal / DFADD	In-the-Wild	ASVspoof
Extra Trees (20 MFCC)	94.97%	53.0%	50.0%
KNN (20 MFCC)	95.23%	52.5%	50.0%
Original MLP (20 MFCC)	95.65%	53.0%	50.0%

Although the models used very different decision rules, their external behavior was similar. This strongly suggested that the dominant limitations were the input representation and the training-domain coverage rather than the choice of classifier alone.

Effect of multi-dataset balancing

Training design	Held-out accuracy	In-the-Wild accuracy	Key observation
DFADD-heavy multi-dataset MLP (24k, 20 features)	95.69%	53.5%	Only a small external gain
Equal-source DFADD + ASVspoof MLP (16k, 20 features)	96.06%	55.0%	Source balance helped modestly
Equal-source richer MLP (16k, 130 features)	99.75%	69.5%	Largest generalization improvement

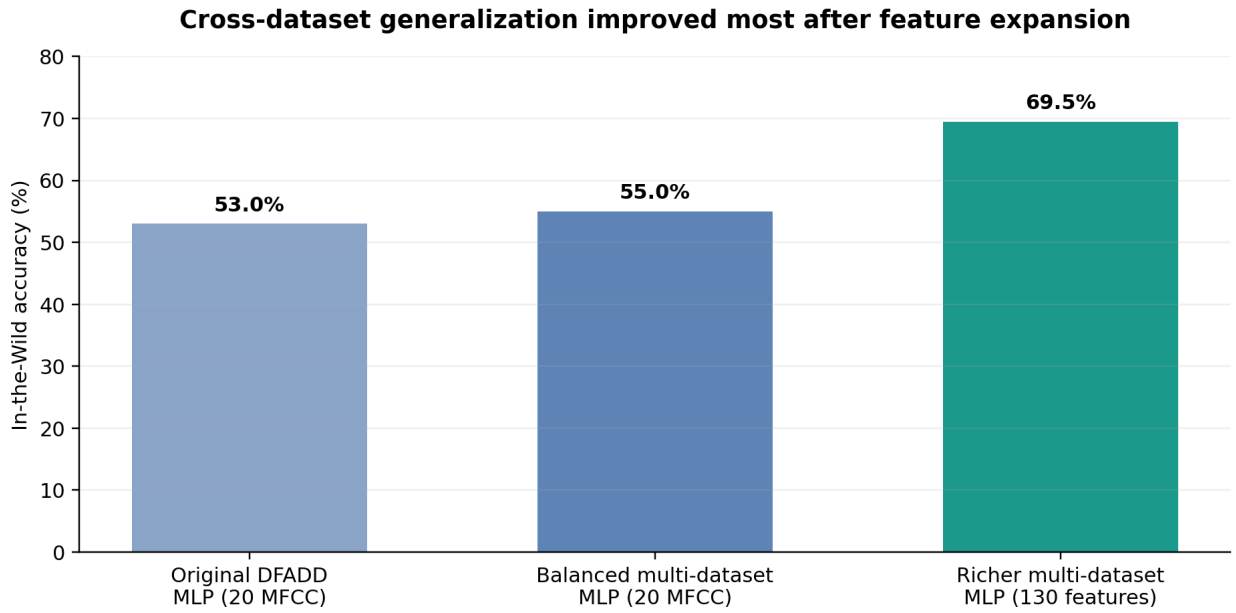


Figure 2. In-the-Wild accuracy improved modestly from source balancing and substantially after richer feature extraction.

Final richer MLP

The final primary model used 8,000 DFADD and 8,000 ASVspoof training examples, balanced by source and class, with 130 acoustic features. On its held-out mixed-source test set, it achieved 99.75% accuracy with only eight errors out of 3,200 examples.

Evaluation	Accuracy	Confusion matrix
Held-out mixed DFADD + ASVspoof	99.75%	[[1595, 5], [3, 1597]]
In-the-Wild (100 real / 100 spoof)	69.50%	[[45, 55], [6, 94]]
ASVspoof evaluation (100 / 100)	81.00%	[[100, 0], [38, 62]]

On In-the-Wild, spoof recall was 94%, while real recall was only 45%. The model therefore remained much better at identifying synthetic speech than at recognizing unfamiliar genuine recordings. The ASVspoof evaluation result was higher, but it is a less independent test because ASVspoof training/development data contributed to the training corpus.

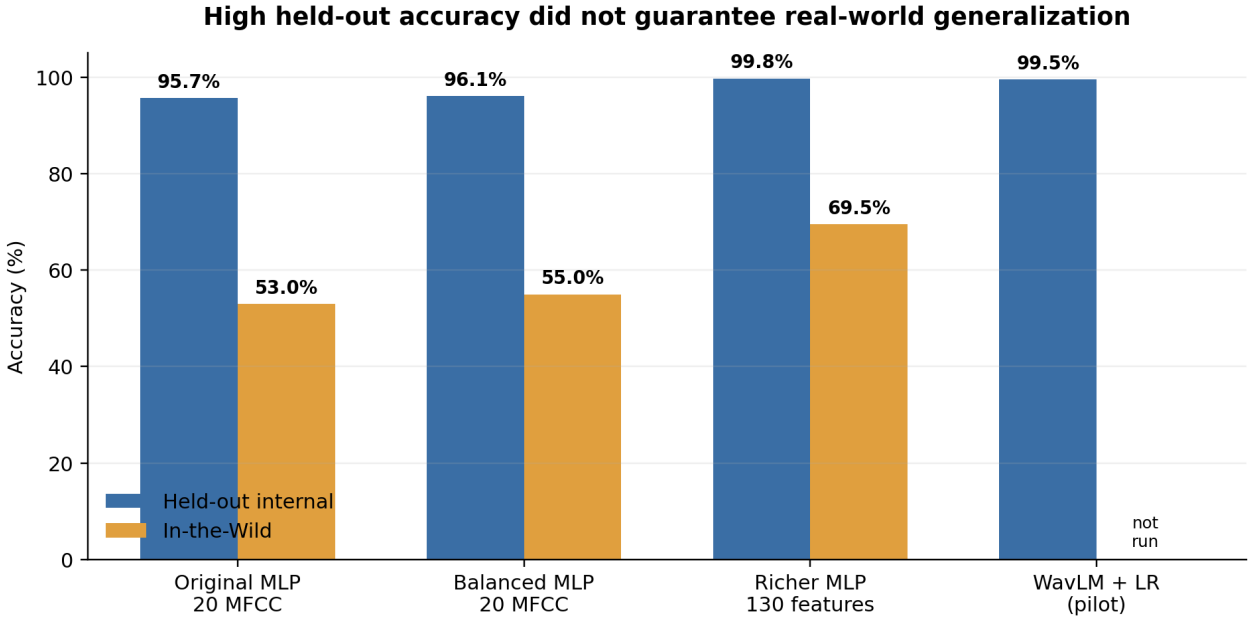


Figure 3. Internal held-out accuracy remained much higher than independent In-the-Wild performance across the project.

Augmentation experiment

The augmented richer MLP used 8,000 feature rows representing 4,000 originals and 4,000 augmented copies. A group-aware split prevented original/augmented pair leakage. Held-out accuracy remained high at 98.50% with confusion matrix [[784, 16], [8, 792]]. However, the first genuine microphone recording was still labeled spoofed with 99.98% confidence. Simple channel/noise augmentation therefore did not close the deployment gap.

Richer Extra Trees and WavLM pilots

Experiment	Held-out result	Local deployment result	Conclusion
Extra Trees on 130 features	99.38%	3/4 labels correct, but ~50–58% confidence	Interesting but too uncertain to replace MLP
WavLM Base Plus + logistic regression (1k pilot)	99.50%	Both real microphone samples mislabeled spoof	Pretrained representation did not automatically remove domain shift

5 Deployment Stress Tests

The most revealing tests used recordings that were not drawn from any benchmark dataset. Two genuine recordings were captured locally using QuickTime / a Mac microphone, and two synthetic recordings were generated using macOS text-to-speech. Because the sample is very small, these tests are qualitative stress tests rather than an accuracy study.

Sample	Ground truth	Richer MLP prediction	Confidence	Correct?
Real recording #1	Real	Spoofed	~100%	No
Real recording #2	Real	Spoofed	99.47% (WAV); 99.36% on website M4A	No
Synthetic recording #1	Synthetic	Spoofed	92.75%	Yes
Synthetic recording #2	Synthetic	Spoofed	100.00%	Yes

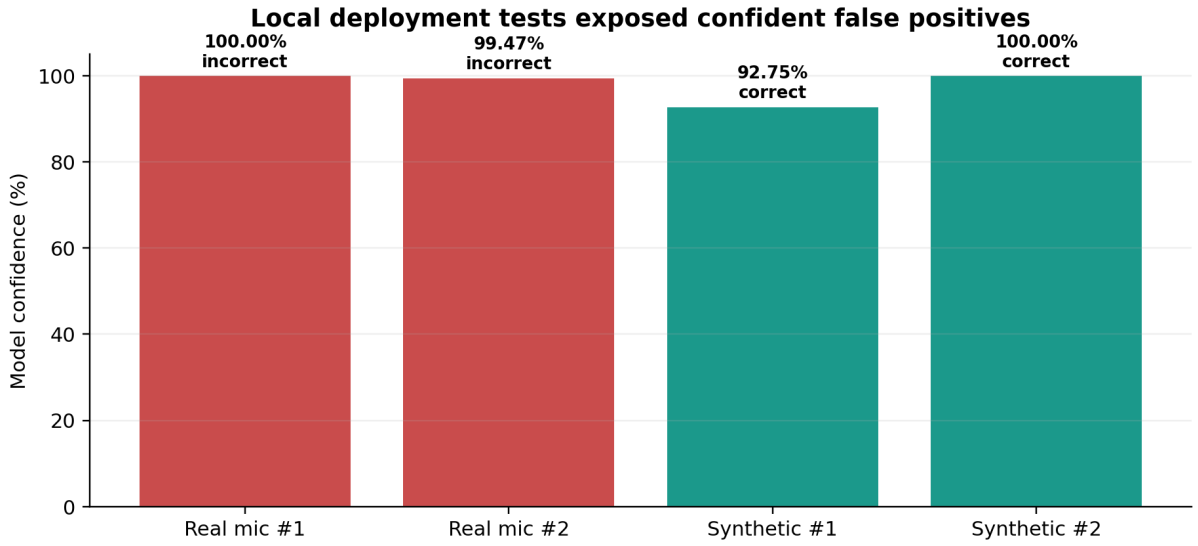


Figure 4. The final model correctly detected the two local synthetic clips but produced high-confidence false positives on genuine microphone recordings.

Why this matters
A model can be confidently wrong. The displayed score is a model output, not a calibrated probability that an audio clip is genuinely fake. The local tests demonstrate why confidence should not be presented as forensic certainty.

Format-conversion check

To determine whether M4A compression alone caused the false positive, one genuine recording was converted to 16 kHz WAV and tested again. The model still labeled it spoofed with extremely high confidence. This indicates that the failure was not explained by file format alone and is more consistent with a broader recording-channel or device-domain mismatch.

6 Discussion

1. Domain shift was the central technical problem

The strongest pattern across all experiments was a large gap between within-distribution and out-of-distribution performance. Models that appeared nearly perfect on held-out samples from familiar datasets could fail close to chance when the evaluation source changed. This is the defining result of the project.

2. Feature representation mattered more than classifier swapping

Extra Trees and KNN did not materially improve In-the-Wild accuracy when they used the same 20-MFCC representation. By contrast, moving to 130 features increased In-the-Wild accuracy from 55.0% to 69.5% under equal-source training. This suggests that retaining variation and temporal-change information was more valuable than simply changing the decision algorithm.

3. Dataset diversity helped, but diversity alone was not enough

Adding ASVspoof to training improved coverage of a second benchmark domain, and equalizing source contributions reduced DFADD dominance. The gain from source balancing alone was modest, however. The remaining error pattern shows that two benchmark datasets still did not represent the full variety of real recording conditions.

4. The detector had asymmetric error behavior

The richer model retained very high spoof recall on In-the-Wild (94%) but only 45% recall for genuine speech. In practical use, this would create a substantial false-positive risk: authentic recordings from unfamiliar environments could be incorrectly flagged as synthetic.

5. High confidence did not imply trustworthy certainty

The local microphone tests are especially important because the model assigned near-100% confidence to wrong predictions. Neural-network probability outputs can be poorly calibrated under distribution shift. A deployment system should therefore distinguish a raw model score from a validated probability of authenticity.

7 Limitations and Ethical Considerations

- The project did not train on the entire ~42.7 GB DFADD corpus; experiments used controlled subsets for computational feasibility.
- The WavLM pilot used only 1,000 clips, so its 99.5% held-out result is not directly comparable to the 16,000-example richer MLP.
- The local real/synthetic stress test used only four recordings and should not be interpreted as a statistical accuracy estimate.
- Only a small range of microphones, rooms, codecs, speakers, and synthesis systems were represented in local testing.
- ASVspoof evaluation data was held out, but the final training set included ASVspoof train/development data from the same corpus family.
- The confidence score was not calibrated and should not be interpreted as the probability that a recording is fake.
- The 130-feature model still summarizes temporal behavior statistically; it does not model the full raw waveform sequence end-to-end.

Ethical implication
Because false positives can wrongly label authentic speech as fabricated, a deepfake detector should be used as supporting evidence rather than definitive proof. The project website therefore describes the system as an experimental research prototype.

8 Final System and Prototype

The final website was connected to the richer 130-feature MLP through a Flask backend. Uploaded audio is saved temporarily, standardized, converted into the same 130-feature representation used during training, classified by the MLP, and then displayed as “Likely Real” or “Likely Synthetic” with a model confidence score and a research disclaimer.

End-to-end verification used the same synthetic file that had already been tested through the command-line predictor. The Python predictor returned “spoofed” at 92.75%, and the website returned “Likely Synthetic” at the same 92.75% score. This confirmed that the web interface was invoking the actual model rather than displaying placeholder results.

9 Conclusion

The project succeeded in building a functional deepfake-audio detection pipeline, but its more important contribution was demonstrating the difference between benchmark success and deployment reliability. The initial balanced DFADD MLP achieved 95.65% held-out accuracy yet only 53% on In-the-Wild and 50% on ASVspoof. Changing classifiers did not solve the problem. Equal-source training produced a modest improvement, while

expanding the acoustic representation to 130 MFCC, delta, delta-delta, and spectral statistics produced the largest gain, raising In-the-Wild accuracy to 69.5% and ASVspoof evaluation accuracy to 81%.

Even after these improvements, genuine microphone recordings could still be classified as synthetic with near-100% confidence. A WavLM pilot also achieved 99.5% held-out accuracy while failing both genuine local microphone tests. These results show that deepfake-audio detection is fundamentally a generalization problem: strong performance on familiar benchmark data does not guarantee reliable behavior under new recording conditions.

Final takeaway
The most defensible conclusion is not that the system is "99% accurate," but that richer representations and broader training improve generalization while substantial domain-shift and calibration problems remain. Cross-dataset and real-device testing are essential before a detector can be trusted outside the laboratory.

10 Future Work

- Collect a larger genuine-speech corpus from phones, laptops, messaging apps, video calls, and varied rooms.
- Expand synthetic training data across more text-to-speech and voice-cloning systems.
- Fine-tune pretrained speech encoders such as WavLM rather than using frozen embeddings only.
- Evaluate end-to-end spectrogram or waveform models that retain temporal structure.
- Add explicit out-of-distribution detection so unfamiliar audio can be labeled "uncertain" rather than forced into real/spoof.
- Calibrate prediction scores using held-out validation data and domain-aware calibration methods.
- Test ensembles and domain-adaptation methods using independent device and codec splits.

11 Comprehensive Experiment Summary

Experiment	Training / features	Internal result	External / deployment result
Early Random Forest baseline	592 balanced DFADD; 20 MFCC	~89.9%	Not externally tested
Balanced DFADD MLP	20k DFADD; 20 MFCC	95.65%	53% ITW; 50% ASV
Extra Trees	20k DFADD features; 20 MFCC	94.97%	53% ITW; 50% ASV
KNN	20k DFADD features; 20 MFCC	95.23%	52.5% ITW; 50% ASV
DFADD-heavy multi-dataset MLP	20k DFADD + 4k ASV; 20 MFCC	95.69%	53.5% ITW
Equal-source multi-dataset MLP	8k DFADD + 8k ASV; 20 MFCC	96.06%	55% ITW
Richer multi-dataset MLP	8k DFADD + 8k ASV; 130 features	99.75%	69.5% ITW; 81% ASV
Augmented richer MLP	4k originals + 4k augmented; 130 features	98.50%	Real mic still spoofed at 99.98%
Richer Extra Trees	16k richer features	99.38%	Local 3/4 labels; weak ~50–58% confidence
WavLM + logistic regression pilot	1k balanced clips; 768-d embeddings	99.50%	Both real mic tests predicted spoof

Historical development note

Earlier progress-report tables listed nominal 1,000 / 10,000 / 20,000 model comparisons with accuracies near 95–99%. We later discovered that the nominal "20,000" collection contained only about 4,031 real examples and 15,969 spoofed examples before balancing, leaving roughly 8,062 examples after class balancing. Those values

remain useful as development history but should not be presented as the final controlled 20,000-sample benchmark. The final balanced DFADD MLP result of 95.65% is the cleaner comparison.

References

[1] DFADD dataset. Hugging Face: isjwdu/DFADD; project repository: isjwdu/DFADD.

[2] SpeechAntiSpoofingBenchmarks/InTheWild dataset. Hugging Face benchmark collection used for independent cross-dataset evaluation.

[3] ASVspoof 2019 Logical Access corpus. Training data accessed through Bisher/ASVspoof_2019_LA and evaluation material through SpeechAntiSpoofingBenchmarks/ASVspoof2019_LA.

[4] Microsoft WavLM Base Plus pretrained speech model. Hugging Face model: microsoft/wavlm-base-plus.

[5] Librosa audio analysis library for resampling and MFCC/spectral feature extraction.

[6] Scikit-learn for scaling, classical machine-learning classifiers, multilayer perceptron training, and evaluation.